

Activity_Explore hypothesis testing

March 17, 2026

1 Activity: Explore hypothesis testing

1.1 Introduction

You work for an environmental think tank called Repair Our Air (ROA). ROA is formulating policy recommendations to improve the air quality in America, using the Environmental Protection Agency's Air Quality Index (AQI) to guide their decision making. An AQI value close to 0 signals “little to no” public health concern, while higher values are associated with increased risk to public health.

They've tasked you with leveraging AQI data to help them prioritize their strategy for improving air quality in America.

ROA is considering the following decisions. For each, construct a hypothesis test and an accompanying visualization, using your results of that test to make a recommendation:

1. ROA is considering a metropolitan-focused approach. Within California, they want to know if the mean AQI in Los Angeles County is statistically different from the rest of California.
2. With limited resources, ROA has to choose between New York and Ohio for their next regional office. Does New York have a lower AQI than Ohio?
3. A new policy will affect those states with a mean AQI of 10 or greater. Will Michigan be affected by this new policy?

Notes: 1. For your analysis, you'll default to a 5% level of significance. 2. Throughout the lab, for two-sample t-tests, use Welch's t-test (i.e., setting the `equal_var` parameter to `False` in `scipy.stats.ttest_ind()`). This will account for the possibly unequal variances between the two groups in the comparison.

1.2 Step 1: Imports

To proceed with your analysis, import `pandas` and `numpy`. To conduct your hypothesis testing, import `stats` from `scipy`.

Import Packages

```
[12]: # Import relevant packages

import pandas as pd
import numpy as np
```

```
from scipy import stats
```

You are also provided with a dataset with national Air Quality Index (AQI) measurements by state over time for this analysis. Pandas was used to import the file `c4_epa_air_quality.csv` as a dataframe named `aqi`. As shown in this cell, the dataset has been automatically loaded in for you. You do not need to download the .csv file, or provide more code, in order to access the dataset and proceed with this lab. Please continue with this activity by completing the following instructions.

Note: For purposes of your analysis, you can assume this data is randomly sampled from a larger population.

Load Dataset

```
[13]: # RUN THIS CELL TO IMPORT YOUR DATA.

aqi = pd.read_csv('c4_epa_air_quality.csv')
```

1.3 Step 2: Data Exploration

1.3.1 Before proceeding to your deliverables, explore your datasets.

Use the following space to surface descriptive statistics about your data. In particular, explore whether you believe the research questions you were given are readily answerable with this data.

```
[14]: # Explore your dataframe `aqi` here:

print(aqi.head(10))

print(aqi.describe(include='all'))

print(aqi['state_name'].value_counts())
```

	Unnamed: 0	date_local	state_name	county_name	city_name \
0	0	2018-01-01	Arizona	Maricopa	Buckeye
1	1	2018-01-01	Ohio	Belmont	Shadyside
2	2	2018-01-01	Wyoming	Teton	Not in a city
3	3	2018-01-01	Pennsylvania	Philadelphia	Philadelphia
4	4	2018-01-01	Iowa	Polk	Des Moines
5	5	2018-01-01	Hawaii	Honolulu	Not in a city
6	6	2018-01-01	Hawaii	Honolulu	Not in a city
7	7	2018-01-01	Pennsylvania	Erie	Erie
8	8	2018-01-01	Hawaii	Honolulu	Honolulu
9	9	2018-01-01	Colorado	Larimer	Fort Collins

		local_site_name	parameter_name \
0		BUCKEYE	Carbon monoxide
1		Shadyside	Carbon monoxide

2	Yellowstone National Park - Old Faithful Snow ...	Carbon monoxide
3	North East Waste (NEW)	Carbon monoxide
4	CARPENTER	Carbon monoxide
5	Kapolei	Carbon monoxide
6	Kapolei	Carbon monoxide
7	NaN	Carbon monoxide
8	Honolulu	Carbon monoxide
9	Fort Collins - CSU - S. Mason	Carbon monoxide

	units_of_measure	arithmetic_mean	aqi
0	Parts per million	0.473684	7
1	Parts per million	0.263158	5
2	Parts per million	0.111111	2
3	Parts per million	0.300000	3
4	Parts per million	0.215789	3
5	Parts per million	0.994737	14
6	Parts per million	0.200000	2
7	Parts per million	0.200000	2
8	Parts per million	0.400000	5
9	Parts per million	0.300000	6

	Unnamed: 0	date_local	state_name	county_name	city_name \
count	260.000000	260	260	260	260
unique	NaN	1	52	149	190
top	NaN	2018-01-01	California	Los Angeles	Not in a city
freq	NaN	260	66	14	21
mean	129.500000	NaN	NaN	NaN	NaN
std	75.199734	NaN	NaN	NaN	NaN
min	0.000000	NaN	NaN	NaN	NaN
25%	64.750000	NaN	NaN	NaN	NaN
50%	129.500000	NaN	NaN	NaN	NaN
75%	194.250000	NaN	NaN	NaN	NaN
max	259.000000	NaN	NaN	NaN	NaN

	local_site_name	parameter_name	units_of_measure	arithmetic_mean \
count	257	260	260	260.000000
unique	253	1	1	NaN
top	Kapolei	Carbon monoxide	Parts per million	NaN
freq	2	260	260	NaN
mean	NaN	NaN	NaN	0.403169
std	NaN	NaN	NaN	0.317902
min	NaN	NaN	NaN	0.000000
25%	NaN	NaN	NaN	0.200000
50%	NaN	NaN	NaN	0.276315
75%	NaN	NaN	NaN	0.516009
max	NaN	NaN	NaN	1.921053

	aqi
count	260.000000

unique	NaN	
top	NaN	
freq	NaN	
mean	6.757692	
std	7.061707	
min	0.000000	
25%	2.000000	
50%	5.000000	
75%	9.000000	
max	50.000000	
California		66
Arizona		14
Ohio		12
Florida		12
Texas		10
New York		10
Pennsylvania		10
Michigan		9
Colorado		9
Minnesota		7
New Jersey		6
Indiana		5
North Carolina		4
Massachusetts		4
Maryland		4
Oklahoma		4
Virginia		4
Nevada		4
Connecticut		4
Kentucky		3
Missouri		3
Wyoming		3
Iowa		3
Hawaii		3
Utah		3
Vermont		3
Illinois		3
New Hampshire		2
District Of Columbia		2
New Mexico		2
Montana		2
Oregon		2
Alaska		2
Georgia		2
Washington		2
Idaho		2
Nebraska		2
Rhode Island		2

```

Tennessee      2
Maine           2
South Carolina  1
Puerto Rico    1
Arkansas        1
Kansas          1
Mississippi     1
Alabama         1
Louisiana       1
Delaware        1
South Dakota    1
West Virginia   1
North Dakota    1
Wisconsin       1
Name: state_name, dtype: int64

```

HINT 1

Consider referring to the material on descriptive statistics.

HINT 2

Consider using `pandas` or `numpy` to explore the `aqi` dataframe.

HINT 3

Any of the following functions may be useful: - `pandas`: `describe()`, `value_counts()`, `shape()`, `head()` - `numpy`: `unique()`, `mean()`

Question 1: From the preceding data exploration, what do you recognize? You have county-level data available for the first hypothesis.

Ohio and New York offer a larger number of observations in this dataset.

1.4 Step 3. Statistical Tests

Before you proceed, recall the following steps for conducting hypothesis testing:

1. Formulate the null hypothesis and the alternative hypothesis.
2. Set the significance level.
3. Determine the appropriate test procedure.
4. Compute the p-value.
5. Draw your conclusion.

1.4.1 Hypothesis 1: ROA is considering a metropolitan-focused approach. Within California, they want to know if the mean AQI in Los Angeles County is statistically different from the rest of California.

Before proceeding with your analysis, it will be helpful to subset the data for your comparison.

```
[15]: # Create dataframes for each sample being compared in your test
```

```
ca_la = aqi[aqi['county_name']=='Los Angeles']  
ca_other = aqi[(aqi['state_name']=='California') & (aqi['county_name']!='Los_↵  
↵Angeles')]
```

HINT 1

Consider referencing the material on subsetting dataframes.

HINT 2

Consider creating two dataframes, one for Los Angeles, and one for all other California observations.

HINT 3

For your first dataframe, filter to `county_name` of Los Angeles. For your second dataframe, filter to `state_name` of California and `county_name` not equal to Los Angeles.

Formulate your hypothesis: Formulate your null and alternative hypotheses:

- H_0 : There is no difference in the mean AQI between Los Angeles County and the rest of California.
- H_A : There is a difference in the mean AQI between Los Angeles County and the rest of California.

Set the significance level:

```
[16]: # For this analysis, the significance level is 5%
```

```
significance_level = 0.05  
significance_level
```

```
[16]: 0.05
```

Determine the appropriate test procedure: Here, you are comparing the sample means between two independent samples. Therefore, you will utilize a **two-sample -test**.

Compute the P-value

```
[17]: # Compute your p-value here
```

```
stats.ttest_ind(a=ca_la['aqi'], b=ca_other['aqi'], equal_var=False)
```

```
[17]: Ttest_indResult(statistic=2.1107010796372014, pvalue=0.049839056842410995)
```

HINT 1

Consider referencing the material on how to perform a two-sample t-test.

HINT 2

In `ttest_ind()`, `a` is the `aqi` column from our “Los Angeles” dataframe, and `b` is the `aqi` column from the “Other California” dataframe.

HINT 3

Be sure to set `equal_var = False`.

Question 2. What is your P-value for hypothesis 1, and what does this indicate for your null hypothesis? Since the p-value (0.049) is less than the significance level of 0.05, we reject the null hypothesis in favor of the alternative. This suggests that adopting a metropolitan strategy may be appropriate in this case.

1.4.2 Hypothesis 2: With limited resources, ROA has to choose between New York and Ohio for their next regional office. Does New York have a lower AQI than Ohio?

Before proceeding with your analysis, it will be helpful to subset the data for your comparison.

```
[29]: # Create dataframes for each sample being compared in your test
```

```
ny = aqi[aqi['state_name']=='New York']  
ohio = aqi[aqi['state_name']=='Ohio']
```

HINT 1

Consider referencing the materials on subsetting dataframes.

HINT 2

Consider creating two dataframes, one for New York, and one for Ohio observations.

HINT 3

For your first dataframe, filter to `state_name` of New York. For your second dataframe, filter to `state_name` of Ohio.

Formulate your hypothesis: Formulate your null and alternative hypotheses:

- H_0 : The mean AQI of New York is greater than or equal to that of Ohio.
- H_A : The mean AQI of New York is **below** that of Ohio.

Significance Level (remains at 5%)

Determine the appropriate test procedure: Here, you are comparing the sample means between two independent samples in one direction. Therefore, you will utilize a **two-sample -test**.

Compute the P-value

```
[32]: # Compute your p-value here
```

```
stats.ttest_ind(a=ny['aqi'], b=ohio['aqi'], alternative='less', equal_var=False)
```

```
[32]: Ttest_indResult(statistic=-2.025951038880333, pvalue=0.030446502691934697)
```

HINT 1

Consider referencing the material on how to perform a two-sample t-test.

HINT 2

In `ttest_ind()`, `a` is the `aqi` column from the “New York” dataframe, and `b` is the `aqi` column from the “Ohio” dataframe.

HINT 3

You can assign `tstat`, `pvalue` to the output of `ttest_ind`. Be sure to include `alternative = less` as part of your code.

Question 3. What is your P-value for hypothesis 2, and what does this indicate for your null hypothesis? With a p-value of 0.030, which is less than the 5% significance level, and a t-statistic of -2.036, we reject the null hypothesis in favor of the alternative. Thus, at the 5% significance level, we can conclude that New York has a lower mean AQI than Ohio.

1.4.3 Hypothesis 3: A new policy will affect those states with a mean AQI of 10 or greater. Will Michigan be affected by this new policy?

Before proceeding with your analysis, it will be helpful to subset the data for your comparison.

```
[33]: # Create dataframes for each sample being compared in your test
```

```
michigan = aqi[aqi['state_name']=='Michigan']
```

HINT 1

Consider referencing the material on subsetting dataframes.

HINT 2

Consider creating one dataframe which only includes Michigan.

Formulate your hypothesis: Formulate your null and alternative hypotheses here:

- H_0 : The mean AQI of Michigan is less than or equal to 10.
- H_A : The mean AQI of Michigan is greater than 10.

Significance Level (remains at 5%)

Determine the appropriate test procedure: Here, you are comparing one sample mean relative to a particular value in one direction. Therefore, you will utilize a **one-sample -test**.

Compute the P-value

```
[37]: # Compute your p-value here
```

```
stats.ttest_1samp(michigan['aqi'], 10 , alternative='greater')
```

```
[37]: Ttest_1sampResult(statistic=-1.7395913343286131, pvalue=0.9399405193140109)
```

HINT 1

Consider referencing the material on how to perform a one-sample t-test.

HINT 2

In `ttest_1samp`), you are comparing the aqi column from your Michigan data relative to 10, the new policy threshold.

HINT 3

You can assign `tstat`, `pvalue` to the output of `ttest_1samp`. Be sure to include `alternative = greater` as part of your code.

Question 4. What is your P-value for hypothesis 3, and what does this indicate for your null hypothesis? Since the p-value (0.940) is greater than the 5% significance level and the t-statistic is -1.74, we fail to reject the null hypothesis.

Therefore, at the 5% significance level, we cannot conclude that Michigan's mean AQI is greater than 10, suggesting that Michigan is unlikely to be affected by the new policy.

1.5 Step 4. Results and Evaluation

Now that you've completed your statistical tests, you can consider your hypotheses and the results you gathered.

Question 5. Did your results show that the AQI in Los Angeles County was statistically different from the rest of California? Yes, the results indicated that the AQI in Los Angeles County was indeed different from that in the rest of California.

Question 6. Did New York or Ohio have a lower AQI? At the 5% significance level, the results indicate that New York has a lower AQI than Ohio.

Question 7: Will Michigan be affected by the new policy impacting states with a mean AQI of 10 or greater? Based on the tests, we fail to reject the null hypothesis, indicating that the mean AQI is not significantly greater than 10. Therefore, Michigan is unlikely to be affected by the new policy.

2 Conclusion

What are key takeaways from this lab? Even with small sample sizes, the variation in the data is sufficient to draw statistically significant conclusions. At the 5% significance level, you found that the mean AQI in Los Angeles was statistically different from the rest of California, and that New York has a lower mean AQI than Ohio. However, you could not conclude that Michigan's mean AQI is greater than 10 at the same significance level.

What would you consider presenting to your manager as part of your findings? For each test, you should present the null and alternative hypotheses, followed by your conclusion and the corresponding p-value that led to it. Since the setup of a t-test involves key configurations that affect result interpretation, you should specify the type of test used, whether it was one-tailed or two-tailed, and how the t-test was performed using the statistical software or library.

What would you convey to external stakeholders? In addressing the research questions, you should report the significance level (5%) and your conclusions. Additionally, including the sample statistics being compared in each case will provide important context, helping stakeholders quickly understand the differences reflected in your results/

Congratulations! You've completed this lab. However, you may not notice a green check mark next to this item on Coursera's platform. Please continue your progress regardless of the check mark. Just click on the "save" icon at the top of this notebook to ensure your work has been logged.